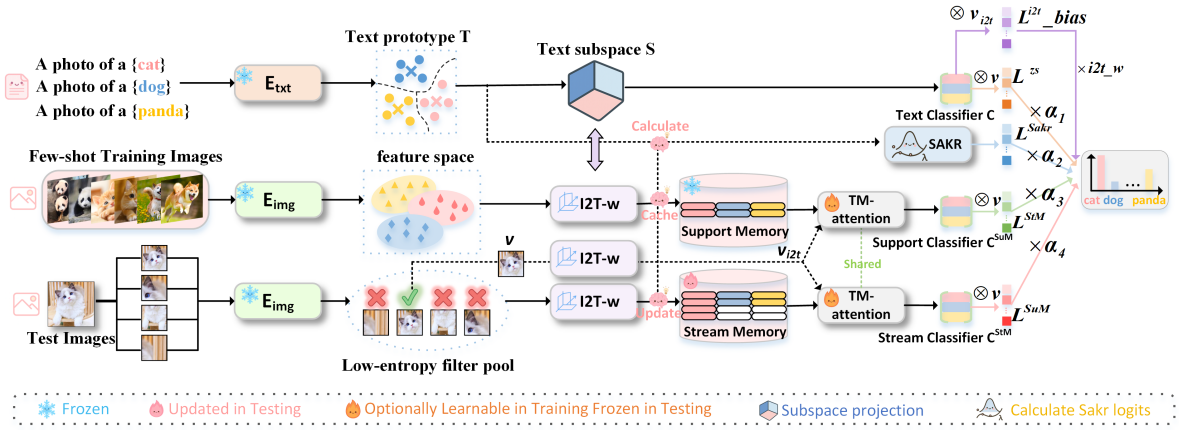
Multi-Bias Logit Fusion in a Pseudo-Text Space for CLIP Few-Shot Learning



🔍 Overview

This repository contains the implementation of MBLF-PTS for image classification with a pre-trained CLIP. We consider four task settings:

Variants	Setting	Stream Memory	Support Memory	SAKR	I2T-w	ω^*
MBLF-PTS-ZS	Zero-shot	✓	✗	✗	✓	✗
MBLF-PTS-TF	Training-free Few-shot	✓	✓	✓	✓	✗
MBLF-PTS	Training-required Few-shot	✓	✓	✓	✓	✓

- Zero-shot classification in a test-time adaptation manner
- Few-shot classification
- Training-free few-shot classification
- Out-of-distribution generalization

📐 Prerequisites

Hardware

This implementation is for the single-GPU configuration. All experiments can be reproduced on a GPU with more than 24GB memory (e.g., 3080Ti)!

Environment

The code is tested on PyTorch 1.13.1.

Datasets

We suggest downloading all datasets to a root directory (`${data_root}`), and renaming the directory of each dataset as suggested in `${ID_to_DIRNAME}` in `./data/datautils.py`. This would allow you to evaluate multiple datasets within the same run.
If this is not feasible, you could evaluate different datasets separately, and change the `${data_root}` accordingly in the bash script.

For zero/few-shot classification, we consider 11 datasets:

- ImageNet
- Flower102
- DTD
- OxfordPets
- StanfordCars
- UCF101
- Caltech101
- Food101
- SUN397
- Aircraft
- EuroSAT

For out-of-distribution generalization, we consider 4 datasets:

- ImageNet-A
- ImageNet-R
- ImageNet-V2
- ImageNet-Sketch

🚀Run MBLF-PTS

We provide a simple bash script under `./scripts/run.sh`. You can modify the paths and other args in the script. One can easily reproduce all results by:

```
bash ./scripts/run.sh
```

zero-shot / training-free few-shot classification:

```
data_root='/root/autodl-tmp/my/data/clip_tuning_datasets/'
testsets=DTD
num_important_channel=250/400/0
lambda_ape=0.3/0.7
lr=0.0001
epoch=20
#arch=RN50
arch=ViT-B/16
bs=32
selection_p=0.1
ctx_init=a_photo_of_a

for nshot in 0
do
  python ./mblf-pts_main.py ${data_root} --test_sets ${testsets} --selection_p
${selection_p}  \
  -a ${arch} -b ${bs}  --ctx_init ${ctx_init}  --memory_size 50 --text_prompt
tip_cupl  --log camera_ready_searched_vit  \
  --gpu 0 --n_shot ${nshot} --n_augview 20   --beta 5.5  --use_searched_param \
  --num_important_channel ${num_important_channel} --lambda_ape ${lambda_ape} --
epoch ${epoch} --lr ${lr}  \
  --use_bridge_branch \
  --bridge_w 0.12
done
```

zero shot generalization to distribution shift:

```
data_root='/root/autodl-tmp/my/data/clip_tuning_datasets/'
testsets=A
num_important_channel=250/400/0
lambda_ape=0.3/0.7
lr=0.0001
epoch=20
#arch=RN50
arch=ViT-B/16
bs=32
selection_p=0.1
ctx_init=a_photo_of_a

for nshot in 0
do
  python ./mblf-pts_main.py ${data_root} --test_sets ${testsets} --selection_p
${selection_p}  \
  -a ${arch} -b ${bs}  --ctx_init ${ctx_init}  --memory_size 50 --text_prompt
tip_cupl  --log camera_ready_searched_vit  \
  --gpu 0 --n_shot ${nshot} --n_augview 20   --beta 8  --use_searched_param \
  --num_important_channel ${num_important_channel} --lambda_ape ${lambda_ape} --
epoch ${epoch} --lr ${lr}  \
  --use_bridge_branch \
  --bridge_w 0.12
done
```

Few-shot classification:

```
data_root='/root/autodl-tmp/my/data/clip_tuning_datasets/'
testsets=DTD
num_important_channel=0
lambda_ape=0.3
lr=0.0001/0.001
epoch=20/100
arch=ViT-B/16
#arch=RN50
bs=32
selection_p=0.1
ctx_init=a_photo_of_a

for nshot in 1
do
  python ./mblf-pts_main.py ${data_root} --test_sets ${testsets} --selection_p
${selection_p}  \
  -a ${arch} -b ${bs}  --ctx_init ${ctx_init}  --memory_size 50 --text_prompt
tip_cupl  --log camera_ready_searched_vit  \
  --gpu 0 --n_shot ${nshot} --n_augview 10   --beta 5.5  --use_searched_param \
  --num_important_channel ${num_important_channel} --lambda_ape ${lambda_ape} --
epoch ${epoch} --lr ${lr} --ft \
  --tda --tda_pos_k 2 --tda_neg_k 1 --tda_pos_beta 5.0 --tda_neg_beta 4.5 \
  --tda_neg_entropy_low 0.3 --tda_neg_entropy_high 0.55 --tda_neg_mask_low 0.03
--tda_neg_mask_high 0.30 \
  --use_sakr_branch --sakr_beta 4.0 --sakr_lambda 1.6 \
```

```
  --use_bridge_branch \
  --bridge_w 0.18
done
```

More shot settings:

```
for nshot in 2
do
  python ./dmn_main.py ${data_root} --test_sets ${testsets} --selection_p
${selection_p}  \
  -a ${arch} -b ${bs}  --ctx_init ${ctx_init}  --memory_size 50 --text_prompt
tip_cupl  --log camera_ready_dmn_searched_vit  \
  --gpu 0 --n_shot ${nshot} --n_augview 20   --beta 5.5  --use_searched_param \
  --num_important_channel ${num_important_channel} --lambda_ape ${lambda_ape} --
epoch ${epoch} --lr ${lr} --ft \
  --tda --tda_pos_k 3 --tda_neg_k 1 --tda_pos_beta 5.5 --tda_neg_beta 4.8 \
  --tda_neg_entropy_low 0.28 --tda_neg_entropy_high 0.55 --tda_neg_mask_low 0.03
--tda_neg_mask_high 0.30 \
  --use_proker_branch --proker_beta 5.0 --proker_lambda 1.4 \
  --use_bridge_branch \
  --bridge_w 0.22 \
  --bridge_train --bridge_epochs 20 --bridge_lr 0.001 --bridge_weight_decay 0.01
done

for nshot in 4
do
  python ./dmn_main.py ${data_root} --test_sets ${testsets} --selection_p
${selection_p}  \
  -a ${arch} -b ${bs}  --ctx_init ${ctx_init}  --memory_size 50 --text_prompt
tip_cupl  --log camera_ready_dmn_searched_vit  \
  --gpu 0 --n_shot ${nshot} --n_augview 20   --beta 5.5  --use_searched_param \
  --num_important_channel ${num_important_channel} --lambda_ape ${lambda_ape} --
epoch ${epoch} --lr ${lr} --ft \
  --tda --tda_pos_k 4 --tda_neg_k 2 --tda_pos_beta 6.0 --tda_neg_beta 5.0 \
  --tda_neg_entropy_low 0.25 --tda_neg_entropy_high 0.55 --tda_neg_mask_low 0.03
--tda_neg_mask_high 0.30 \
  --use_proker_branch --proker_beta 6.0 --proker_lambda 1.2 \
  --use_bridge_branch \
  --bridge_w 0.25 \
  --bridge_train --bridge_epochs 20 --bridge_lr 0.001 --bridge_weight_decay 0.01
done

for nshot in 8
do
  python ./dmn_main.py ${data_root} --test_sets ${testsets} --selection_p
${selection_p}  \
  -a ${arch} -b ${bs}  --ctx_init ${ctx_init}  --memory_size 50 --text_prompt
tip_cupl  --log camera_ready_dmn_searched_vit  \
  --gpu 0 --n_shot ${nshot} --n_augview 24   --beta 5.5  --use_searched_param \
  --num_important_channel ${num_important_channel} --lambda_ape ${lambda_ape} --
epoch ${epoch} --lr ${lr} --ft \
  --tda --tda_pos_k 7 --tda_neg_k 3 --tda_pos_beta 6.5 --tda_neg_beta 5.2 \
  --tda_neg_entropy_low 0.25 --tda_neg_entropy_high 0.55 --tda_neg_mask_low 0.03
--tda_neg_mask_high 0.30 \
  --use_proker_branch --proker_beta 7.5 --proker_lambda 1.0 \
```

```
  --use_bridge_branch \
  --bridge_w 0.28 \
  --bridge_train --bridge_epochs 20 --bridge_lr 0.001 --bridge_weight_decay 0.01
done

for nshot in 16
do
  python ./dmn_main.py ${data_root} --test_sets ${testsets} --selection_p
${selection_p}  \
  -a ${arch} -b ${bs}  --ctx_init ${ctx_init}  --memory_size 50 --text_prompt
tip_cupl  --log camera_ready_dmn_searched_vit  \
  --gpu 0 --n_shot ${nshot} --n_augview 64   --beta 5.5  --use_searched_param \
  --num_important_channel ${num_important_channel} --lambda_ape ${lambda_ape} --
epoch ${epoch} --lr ${lr} --ft \
  --tda --tda_pos_k 8 --tda_neg_k 3 --tda_pos_beta 7.0 --tda_neg_beta 5.5 \
  --tda_neg_entropy_low 0.22 --tda_neg_entropy_high 0.55 --tda_neg_mask_low 0.03
--tda_neg_mask_high 0.30 \
  --use_proker_branch --proker_beta 8.0 --proker_lambda 0.9 \
  --use_bridge_branch \
  --bridge_w 0.30 \
  --bridge_train --bridge_epochs 20 --bridge_lr 0.001 --bridge_weight_decay 0.01
done
```

📜 Main Results

Zero-shot Classification

Method	ImageNet	Flower	DTD	Pets	Cars	UCF	Caltech	Food	SUN	Aircraft	EuroSAT	Mean
CLIP-RN50(Radford et al., 2021)	58.16	61.75	40.37	83.57	55.70	58.84	85.88	73.97	58.80	15.66	23.69	56.04
DiffTPT* (Feng et al., 2023)	60.80	63.53	40.72	83.40	60.71	62.67	86.89	**79.21**	62.72	17.60	41.04	59.94
VisDesc(Menon & Vondrick, 2022)	59.68	65.37	41.96	82.39	54.76	58.47	88.11	76.80	59.84	16.26	37.60	58.29
Ensemble(Zhang et al., 2022)	60.32	66.10	40.07	85.83	55.71	61.33	83.94	77.32	58.53	17.10	37.54	58.53
CALIP(Guo et al., 2023)	60.57	66.38	42.39	86.21	56.27	61.72	87.71	77.42	58.59	17.76	38.90	59.45
CuPL(Pratt et al., 2023)	61.45	65.44	48.64	84.84	57.28	58.97	89.29	76.94	62.55	19.59	38.38	60.31
SuS-X* (Udandarao et al., 2023)	61.84	67.72	50.59	85.34	57.27	61.54	89.53	77.58	62.95	19.47	45.57	61.76
DMN(Zhang et al., 2024)	**63.87**	67.93	50.41	86.78	60.02	65.34	90.14	76.70	64.39	**22.77**	48.72	63.37
TDA(Karmanov et al., 2024)	61.35	68.74	43.74	86.18	57.78	64.18	89.70	77.75	62.53	17.61	42.11	61.06
ZLaP† (Kalantidis et al., 2024)	62.20	69.27	42.79	80.32	56.42	62.81	86.90	77.87	61.83	17.37	31.85	59.06
ECALP(Li et al., 2025b)	62.64	69.39	**54.49**	**88.20**	60.56	66.67	89.94	76.97	64.97	21.12	49.09	64.00
ADAPT(Zhang et al., 2025)	62.16	70.04	51.89	86.43	58.38	63.12	89.37	75.57	64.94	18.00	50.47	62.76
MBLF-PTS-ZS (Ours)	63.84	**71.01**	52.90	87.16	**60.71**	**66.93**	**90.67**	76.83	**65.89**	22.23	**50.17**	**64.39**
CLIP-ViT-B/16(Radford et al., 2021)	66.73	64.44	44.27	88.25	65.48	65.13	93.35	83.65	62.59	23.67	42.01	63.87
DiffTPT* (Feng et al., 2023)	70.30	70.10	47.00	88.20	67.01	68.22	92.49	**87.23**	65.74	25.60	43.13	65.90
DMN(Zhang et al., 2024)	**72.25**	74.49	55.85	92.04	67.96	72.51	**95.38**	85.08	70.18	30.03	59.43	70.47
TDA(Karmanov et al., 2024)	69.51	71.42	47.40	88.63	67.28	70.66	94.24	86.14	67.62	23.91	58.00	67.71
ZLaP† (Kalantidis et al., 2024)	70.17	73.49	48.58	87.14	65.63	71.45	93.06	86.92	67.44	25.44	55.62	67.72
ECALP(Li et al., 2025b)	71.26	75.96	56.32	**92.31**	68.20	75.44	94.40	85.72	70.35	29.49	56.53	70.54
ADAPT(Zhang et al., 2025)	70.91	75.56	55.20	92.01	68.19	70.66	94.48	83.81	70.57	28.95	**68.19**	70.78
MBLF-PTS-ZS (Ours)	72.23	**77.30**	**56.80**	92.04	**68.76**	**75.68**	95.13	85.39	**71.41**	**30.84**	59.77	**71.40**

Few-shot Classification

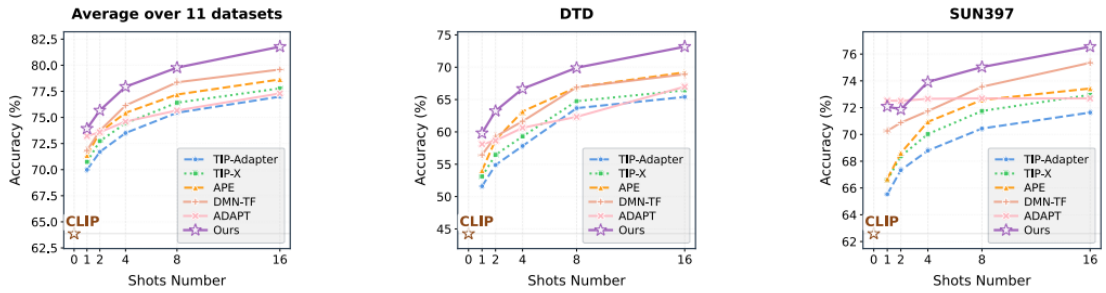


Figure 4. Training-free few-shot results with ViTB/16 backbone. Full results on 11 classification datasets are presented in Fig. 12.

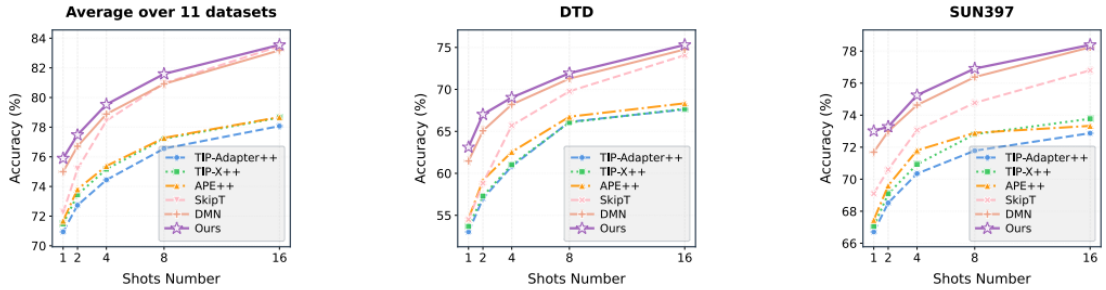
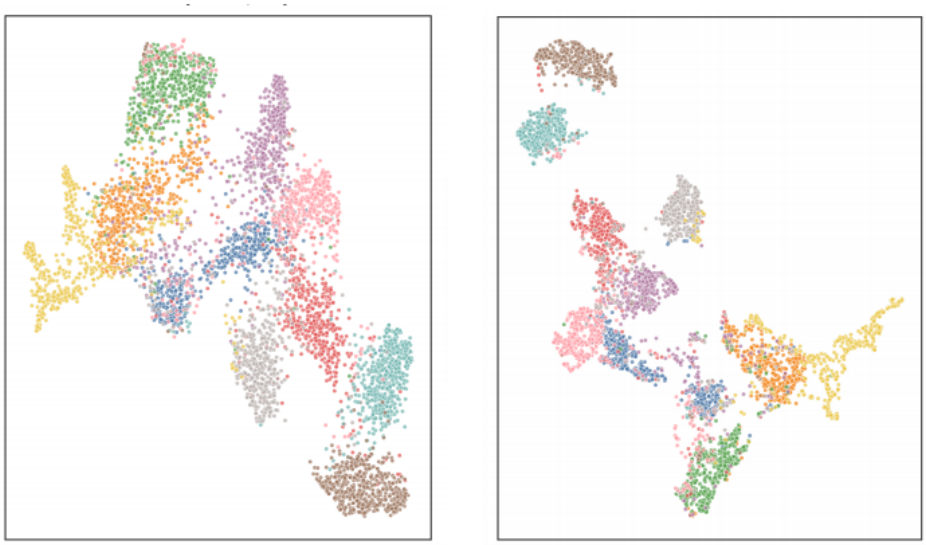


Figure 5. Few-shot performance with ViTB/16 backbone, where the full results on 11 classification datasets are presented in Fig. 13.

Out-of-Distribution Generalization

Method	ImageNet Variants			
	-A	-V2	-R	-S
CLIP-RN50(Radford et al., 2021)	21.83	51.41	56.15	33.37
TPT(Manli et al., 2022)	26.67	54.70	59.11	35.09
CALIP(Guo et al., 2023)	23.96	53.70	60.81	35.61
DiffTPT(Feng et al., 2023)	**31.06**	55.80	58.80	37.10
CoCoOp*(Zhou et al., 2022)	23.32	55.72	57.74	34.48
CoOp*(Zhou et al., 2022)	23.06	55.40	56.60	34.67
DMN(Zhang et al., 2024)	28.57	56.12	61.44	**39.84**
MBLF-PTS (Ours)	28.95	**56.41**	**61.82**	39.81
CLIP-ViT(Radford et al., 2021)	47.87	60.86	73.98	46.09
TPT(Manli et al., 2022)	54.77	63.45	77.06	47.94
MaPLe*(Khattak et al., 2023a)	50.90	64.07	76.98	49.15
PromptSRC*(Khattak et al., 2023b)	50.90	64.35	77.80	49.55
GDA(Wang et al., 2024)	19.72	55.67	55.30	34.32
TIMO*(Li et al., 2025a)	22.06	56.40	58.47	35.96
DMN(Zhang et al., 2024)	58.28	65.12	78.55	53.02
MBLF-PTS (Ours)	**58.80**	**65.15**	**79.20**	**53.26**

I2T-w Visualization



(a) Zero-shot *(b)* I2T-w